Introduction to Trigonometry

The big idea

Trigonometry links the **angles** of a right triangle to the **ratios of its sides**. Once you know one acute angle and one side, you can work out every other side.

The six ratios

In right △ABC with the right angle at B, for the acute angle A: the side opposite A is BC, the side next to A (adjacent) is AB, and the hypotenuse is AC.

Ratio	Definition	Memory
sin A	opposite / hypotenuse	**S**ome **P**eople **H**ave
cos A	adjacent / hypotenuse	**C**urly **B**lack **H**air
tan A	opposite / adjacent	**T**urning **P**ermanently **B**lack
cosec A	1 / sin A	
sec A	1 / cos A	
cot A	1 / tan A	

Also **tan A = sin A / cos A**. The ratios depend only on the angle, not on the size of the triangle.

IF TAN A = 4/3, FIND SIN A AND COS A

Take opposite = 4k, adjacent = 3k. Hypotenuse = $\sqrt{(16k^2 + 9k^2)} = 5k$.
sin A = 4/5, cos A = 3/5.

Values at specific angles

	0°	30°	45°	60°	90°
sin	0	1/2	1/√2	√3/2	1
cos	1	√3/2	1/√2	1/2	0
tan	0	1/√3	1	√3	not defined

REMEMBER JUST ONE ROW

sin 0°, 30°, 45°, 60°, 90° = √0/2, √1/2, √2/2, √3/2, √4/2. Cos is the same row backwards, and tan = sin/cos.

Trigonometric identities

THE THREE IDENTITIES (FOR 0° ≤ A ≤ 90°)

$$\sin^2 A + \cos^2 A = 1$$

$$1 + \tan^2 A = \sec^2 A$$

$$1 + \cot^2 A = \operatorname{cosec}^2 A$$

PROVE (1 − COS²A) COSEC²A = 1

$1 - \cos^2 A = \sin^2 A$, and $\sin^2 A \times \operatorname{cosec}^2 A = \sin^2 A \times 1/\sin^2 A = \mathbf{1}$.

SIN²A IS NOT SIN(A²)

$\sin^2 A$ means $(\sin A)^2$. And $\sin(A + B)$ is **not** $\sin A + \sin B$.

STRATEGY FOR IDENTITY PROOFS

Start from the more complicated side. Convert everything to sin and cos, use $\sin^2 A + \cos^2 A = 1$, and simplify.

Quick recap

- sin = O/H, cos = A/H, tan = O/A; cosec, sec, cot are their reciprocals.
- Learn the table for 0°, 30°, 45°, 60°, 90°.
- Three identities, all from Pythagoras.

Practice questions

1. If sin A = 1/2, then A equals
   (a) 60° (b) 30° (c) 45° (d) 90°
2. The value of tan 45° + cos 60° is
   (a) 1 (b) 3/2 (c) 2 (d) √3
3. $\sec^2 A - \tan^2 A$ equals
   (a) 0 (b) 1 (c) −1 (d) 2
4. If cos A = 4/5, then tan A is
   (a) 3/4 (b) 4/3 (c) 3/5 (d) 5/3
5. Which of these is not defined?
   (a) sin 90° (b) cos 0° (c) tan 90° (d) cot 45°

Answers: 1. b 2. b 3. b 4. a 5. c